

From "Change" to Everything

— A Popular Science Introduction to the Coarse-Graining Genesis Framework V4.0

You look up at the stars and the universe seems quiet and empty. But a physicist will tell you: inside the "empty nothingness" all around you, countless particles are popping into and out of existence every single instant, with enough energy to boil an ocean.

So here is the question — **how does this frantic change manage, so reliably, to become the apple on your desk, the Moon in the sky, and that invisible "gravity" that pulls the apple to the ground?**

This introduction is an attempt to answer that question. And the answer begins with a claim that sounds very grand: **spacetime does not exist, and gravity is not "already there" — both of them emerge from "change."**

What this document is trying to do

Our two papers — *The spectrum of a compact internal space. I. Gauge structure and fermion content*, and *...II. Effective couplings and mass scales* — together with the "Coarse-Graining Genesis Framework V4.0" that carries their computation, all tell a single story: **the familiar world around us — spacetime, gravity, matter, dark energy — is not "already there"; it all emerges from something more fundamental, namely "change."**

That sentence sounds grand, even a little mystical. The purpose of this introduction is to tell that story in the plainest language possible: why we think this way, what difficulty we ran into, how we solved it, and what we finally computed.

No prior knowledge of quantum field theory is assumed. Wherever something technical is needed, I will first offer an analogy, then add one rigorous sentence. The analogy helps you grasp the intuition; the rigorous sentence keeps me from misleading you.

This introduction follows two parallel threads, and the reader is invited to keep them in mind throughout.

The first thread is about "space": spacetime does not exist, yet the path-integral measure of standard quantum field theory cannot be defined without it. The way out is to change the analysis domain — from "fields in spacetime" to "the spectrum of change" — much as signal processing moves a problem from the time domain to the frequency domain (Chapters 2 and 4).

The second thread is about “field”: constructing the field directly is hard and divergent everywhere — the difficulty of the Yang–Mills conjecture. The way out is to reverse the starting point: change is the substance and the field is derived; and the smallest, most fundamental change is pinned down uniquely by a single axiom, which we call the disorder axiom (Chapters 3 and 6).

1. The Spacetime Dilemma: A Century-Long Conflict

Two great edifices, two spacetimes

Twentieth-century physics erected two magnificent edifices.

One is called the **Standard Model**. It describes the behavior of elementary particles in the microscopic world — quarks, electrons, photons, gluons — and three of the forces between them (electromagnetic, weak, strong). The Standard Model is extraordinarily successful, accurate to more than ten decimal places, one of the most thoroughly verified theories in the history of science.

The other is called **General Relativity**. It describes gravity in the macroscopic world — stars, galaxies, black holes, the evolution of the whole universe. It is equally successful, having predicted the bending of light, gravitational waves, and cosmic expansion, all confirmed by observation.

The problem is that these two edifices hold **fundamentally contradictory** views of “spacetime.”

General Relativity says: spacetime is a **curved, elastic, living thing**. Matter tells spacetime how to curve; spacetime tells matter how to move. Spacetime is itself a dynamical object — it can be bent by mass, it can ripple (gravitational waves), and it can have a beginning and an end (inside a black hole).

The Standard Model says: spacetime is just a **fixed, rigid background stage**. Particles and fields dance on this stage, and the stage itself does not move at all. It assumes a flat, uniform spacetime (physicists call it “Minkowski spacetime”), like an infinite, never-changing grid.

One says spacetime is alive; the other says spacetime is dead. These two things cannot both be true.

First, go deeper: what actually is gravity?

Before discussing the conflict, we need to seriously ask a deceptively simple question: **what, exactly, is gravity?**

Because our answer to “what is gravity” directly determines our attitude toward “what is spacetime.”

The traditional answer (General Relativity): gravity is not a "force" but the curvature of spacetime. The Earth orbits the Sun not because an invisible rope pulls it, but because the Sun "bends" the spacetime around it, and the Earth simply follows a "straight line" (a geodesic) in that curved spacetime. The falling apple, the bending of light, the black hole — all are manifestations of spacetime curvature.

This answer is beautiful, but it binds "gravity" to "the geometry of spacetime": **gravity = spacetime curvature**. And so "spacetime" must be a real thing that can be bent.

Another answer (the quantum field view): gravity is carried by a particle called the "graviton," just as electromagnetism is carried by the photon. The graviton is a massless, spin-2 particle.

But this answer also has trouble: the graviton merely "quantizes gravity as if it were an ordinary force"; it does not answer the deeper question of *why* gravity exists, and its calculations diverge everywhere.

We give a **third answer** — the one running through this entire introduction: **gravity is neither spacetime curvature nor a fundamental force; it is an "emergent" phenomenon — just like temperature.**

How temperature "emerges"

To understand "gravity emerges like temperature," we must first explain what temperature is. This may be the single most important analogy in this document, so please read it carefully.

A cup of water consists of roughly 10^{23} water molecules. Each molecule is in random thermal motion — some fast, some slow, in every direction.

Now ask: **what is the "temperature" of one particular water molecule in this cup?**

The question is meaningless. **A single molecule has no temperature.** You cannot point at one molecule and say "its temperature is 30 degrees."

"Temperature" is a property that belongs only to the **collective of many molecules**:

- The greater the **average kinetic energy** of the molecules, the higher the temperature of the water;
- More precisely, temperature is defined through entropy: $1/T = \partial S / \partial E$ — temperature is the reciprocal of "how much the entropy (disorder, number of states) increases when you add one unit of energy";
- The higher the temperature, the less the entropy increases for the same added energy (the system is already disordered enough not to care).

So the essence of temperature is this: **it lives in no single molecule; it "arises" from the collective motion of billions upon billions of molecules.** Microscopically there is only "molecules moving"; macroscopically there is "temperature" — this arising from micro to macro is what physics calls **emergence**.

Emergence has a distinctive feature: **you cannot find it microscopically, yet macroscopically it exists truly and lawfully**. Cut open a cup of water and you will never find a "temperature" in there; but put your hand in and you will truly feel it.

Wait a moment — "emergence" is not a metaphor; it is mathematically proven

You might wonder: is the claim "temperature is emergent" just a nice metaphor?

Yu Deng, a 2026 Fields Medalist, answered this question with mathematics.

In plain terms, Deng's work proves: **a room full of chaotically bouncing marbles (microscopic particles) really does "emerge" into temperature, pressure, and fluid motion (macroscopic laws)**. This is not philosophical musing; it is a rigorous mathematical proof — starting from Newtonian mechanics and, through extremely intricate derivation, ending at the equations that describe the flow of a gas.

This means: **"emergence" is not a vague concept but a mathematical fact that can be verified and computed**. Temperature is not a property "imposed on" molecules from outside; it is a structure that forms spontaneously from the collective motion of molecules. Gravity is the same — not a property "imposed on" the vacuum, but a structure that forms spontaneously from the collective motion of vacuum fluctuations.

Deng proved that the microscopic can emerge into the macroscopic. **What our framework aims to prove is: change can emerge into spacetime.**

Gravity emerges just like temperature

Now replace "molecules" with "vacuum fluctuations" and "temperature" with "gravity" — the logic is exactly the same.

The vacuum, in quantum field theory, is **not empty**. It is filled with violent "quantum fluctuations" — virtual particle–antiparticle pairs constantly popping into existence and instantly annihilating, like an ever-boiling, invisible soup. Every birth and death of a virtual pair is one microscopic "change."

These fluctuations are not idle: they carry energy and momentum. Physicists describe "how much energy-momentum there is and how it is distributed" with a quantity called the **energy-momentum tensor**.

And the energy-momentum tensor can be further decomposed into its components at each frequency — that is its **spectral function**. The spectral function tells us: among the vacuum fluctuations, how many are low-frequency, how many are high-frequency, and how much energy-momentum each contributes.

Now ask a question exactly parallel to the temperature one: **what is the "gravity" of one particular virtual particle in this soup?**

Again, meaningless. **A single fluctuation has no gravity.** You cannot point at one fleeting virtual particle and say "its gravity is this much."

"Gravity" is a property belonging only to the **collective of all vacuum fluctuations**:

- It is a **collective emergent structure** of the spectral function of the energy-momentum tensor — specifically, that **most stable, most unchanging component (the zero mode)** of the spectrum;
- Just as temperature is the "most averaged, most collective" part of molecular motion, gravity is the "most stable, most unchanging" part of vacuum fluctuations.

So the essence of gravity is this: **it lives in no single fluctuation; it "arises" from the spectral structure of the vast sea of vacuum fluctuations.** Microscopically there is only "fluctuations popping and annihilating"; macroscopically there is "gravity" — this is the emergence of gravity.

Pin the analogy down in one sentence:

> **Temperature is the emergence of molecular motion; gravity is the emergence of vacuum fluctuations.**

Temperature lets us believe that emergence can produce genuinely real macroscopic effects (boiling water will scald you); gravity is the same — it genuinely makes apples fall and planets orbit, but it is not a "fundamental force"; it is the phenomenon that emerges from the spectral structure of vacuum fluctuations.

So what, then, did Einstein's "curved spacetime" actually curve?

Einstein has a famous dictum: "matter tells spacetime how to curve; spacetime tells matter how to move." His gravitational equations are extraordinarily precise, confirmed by observation countless times — we fully respect this.

But if spacetime does not exist, then what exactly is it that Einstein's "curvature" curves?

Our answer: **what curves is the vacuum fluctuation.**

Matter (mass, energy) really does change its surroundings — but what it changes is not "spacetime" (for spacetime is not a thing at all), but the **spectral structure of the vacuum fluctuations**. Where mass gathers, the distribution of vacuum fluctuations (the spectral function of the energy-momentum tensor) changes; this "change in the fluctuation spectrum" is precisely the phenomenon Einstein observed and precisely described.

If we adopt a different ontological viewpoint, the physical picture behind the equations can be another one: **what curves is, in fact, the vacuum fluctuation.** This is not to say relativity no longer works — the phenomena the equations describe and the predictions they make all still stand; rather, we gain a new understanding of the *root* of "curvature": what really "curves" is the spectral structure of the vacuum fluctuations, not the entity called "spacetime."

An analogy: you see a whirlpool on a river and say "the riverbed must be dented here." One can also look at it another way: the riverbed does not exist at all; what actually changes is the water itself. Einstein's equations precisely describe the "whirlpool," simply attributing it to a "dented riverbed"; and we may attribute it differently — what really "curves" and "changes" is the water (vacuum fluctuations), not the riverbed (spacetime).

Space is continuous, but that does not mean it is "simple"

The work of another 2026 Fields Medalist — Hong Wang — touches the essence of "space" from a completely different angle.

Wang proved a problem that had troubled mathematics for a century: the **three-dimensional Kakeya conjecture**. The conjecture asks: a needle that is infinitely thin can point in every direction within a space — how small can that space be?

The answer is surprising: in three dimensions, **there is a positive minimum volume**. You cannot compress "a needle that can point in every direction" down to arbitrarily small size.

What does this mean? **It means "continuous space" is not a monolithic, simple concept**. At microscopic scales, the structure of continuous space can become extremely intricate, extremely "non-smooth." Wang's work reveals: **between the continuous and the discrete, there is a far richer relation than we imagined**.

There is a beautiful symmetry here: **Deng studies "how a crowd of particles forms a continuous field"; Wang studies "how a wave field reveals grain-like structure."** One goes from the discrete to the continuous, the other from the continuous to the discrete — **together they tell us: the continuous and the discrete are not an either/or opposition, but two faces of one world at different scales**.

Our framework goes one step further: **even "space" itself emerges from the spectrum of "change."** Wang proved that continuous space can be very complex; we say that even the concept "continuous" is itself derivative.

Wang's work points, unintentionally, to a deeper question: **if "continuous space" can be this complex at the microscopic level, how can it be "fundamental"? A fundamental thing should be simple. The very complexity of space shows that it is not a source but a result — something that emerges from something simpler.** That simpler thing, in our framework, is "change."

The concrete manifestations of the conflict

With the "gravity is emergent" view in hand, the conflict between the two edifices becomes clearer.

This conflict is not ivory-tower chatter; it "detonates" in concrete problems:

- **The quantum gravity problem:** when we try to merge quantum mechanics with General Relativity into "quantum gravity," the calculations blow up into infinities that cannot be cancelled. The root cause is that General Relativity takes "gravity = spacetime curvature" as its starting point, and the object "curved spacetime" simply cannot stand at the scale of quantum fluctuations.
- **The black hole information paradox:** black holes evaporate (Hawking radiation), so where does the information that fell in go? General Relativity says spacetime terminates at a singularity; quantum mechanics says information cannot simply vanish. The two answers collide.
- **The cosmological constant problem:** the vacuum energy density predicted by theory and that observed differ by 120 orders of magnitude — the most preposterous "error" in the history of physics, its root likewise pointing to the assumption of a "spacetime background" itself.

Our answer: spacetime does not exist

For a hundred years, people have defaulted to "spacetime exists; we simply haven't found the way to unify it yet."

We offer a more radical view: **spacetime does not "exist."**

Spacetime is not a "thing"; it is a **description of change**. It has only two ingredients:

- **Time** = the **ordering** of change (which change comes before which);
- **Space** = the **reach** of change (which changes can influence each other, and which cannot).

An analogy: the water in a river is the "change," the riverbed is "spacetime." Traditional physics holds that the riverbed is real and the water flows on it. We say: **there is no riverbed, only the flowing water**. The so-called "riverbed" is merely a way of describing how the water flows (first or later, and where toward).

General Relativity errs by treating the "curved riverbed" (curved spacetime) as a real thing. The Standard Model errs by treating the "rigid riverbed" (background spacetime) as a real thing. They argue over "is the riverbed curved or flat," and we say — **stop arguing; there is no riverbed at all**.

This one sentence dissolves the conflict between the two edifices at once: their disagreement is not about "the shape of spacetime," but about a mistake they both share — treating "the way we describe change" as "a thing that exists independently."

And gravity? Under the "spacetime does not exist" picture, gravity naturally finds its place: **gravity is not "the curvature of spacetime" (because spacetime is not a thing to bend), but a phenomenon emerging from "the spectral structure of vacuum fluctuations" (the most stable component of change)**. Temperature needs no "heat-riverbed" to carry it; gravity needs no "curved spacetime" to carry it — both are the emergence of change.

2. The Foundation Problem of Quantum Field Theory: Where Does the Measure Come From?

How quantum field theory is "assembled"

The Standard Model is a **quantum field theory**. Quantum field theory has a standard assembly procedure, which can be roughly divided into three steps:

1. **Define the field:** first assume that at every point of space there is a "field" (the electromagnetic field, the electron field, ...). The field is a quantity pervading spacetime.
2. **Write down the path integral:** to compute the probability for "a particle to go from here to there," physicists sum over all possible "paths" (histories), each path contributing a weight. This is Feynman's path integral.
3. **Provide the measure:** this act of "summing over all paths" needs a "ruler" to measure different paths — which path counts as one unit, and how it is weighted. This ruler is mathematically called the **measure**.

The measure is the foundation, and the foundation depends on spacetime

The third step is the key, and the hardest.

What is the measure? It is a metric on the infinite-dimensional space of "all possible field configurations." And what is a "field configuration"? It is a quantity **defined at every point and every instant of spacetime**.

That is: **the measure inherently depends on spacetime**. To measure a field, you must first lay down a grid over spacetime and integrate the field's values at every lattice point. Spacetime is the stage on which the measure is defined.

How hard this step is, one can feel it: the path integral sums over "infinitely many possible field configurations," and each configuration is "values at infinitely many points of spacetime." That is an "infinity of infinities" — two layers of infinity stacked on each other. To this day no one has rigorously and uncontroversially defined this sum. This is not technical laziness; it is a **fundamental difficulty**.

Then the contradiction surfaces

Now bring in the conclusion of Chapter 1:

- spacetime does not exist (it is only a description of change);
- yet the measure of quantum field theory must depend on spacetime to be defined.

Does the measure still have a foundation? **No**.

It is like this: to measure the flow of a river, you must first assume a riverbed exists; but we have just proven the riverbed does not exist. Then with what do you measure the "flow"?

This is the deepest "foundation problem" of quantum field theory: **it stands on a foundation that does not exist.** The so-called "rigorous mathematical construction of quantum field theory remains incomplete," the so-called "path-integral measure is ill-defined" — all these unresolved difficulties trace back to one and the same fact — **the measure needs spacetime, and spacetime does not exist.**

This is also why patching inside quantum field theory can never fill the hole. You are patching "the shape of the riverbed," while the hole is "the riverbed itself." As long as you stand on the assumption "spacetime is the background," the measure remains forever suspended.

Our way out: not a measure, but a spectrum

Since the foundation of the measure (spacetime) does not exist, our strategy is clear: **bypass the measure.**

Do not define "fields on spacetime," do not write "path integrals on spacetime," do not seek "measures on spacetime" — instead start from "the spectrum of change" (the spectrum needs no spacetime background; it needs only "change" itself), and use the spectrum to dually rebuild everything.

This route bypasses the "measure depends on spacetime" dead end. How specifically to do it is the content of Chapters 4 and 5. Here one only needs to remember a chain of logic: **spacetime does not exist → the measure has no foundation → we must bypass the measure → start from the spectrum.**

Changing the analysis domain: from the "time domain" to the "frequency domain"

This chain of logic has a familiar parallel: signal processing.

A complicated sound signal is a tangled mess on the time axis; but one Fourier transform later, decomposed by frequency, it becomes a handful of clean "pure tones." No information is lost — only the basis has changed: instead of asking "how loud at each moment," one asks "how strong at each frequency." Filtering and compression, painful in the time domain, become startlingly simple in the frequency domain.

Standard quantum field theory is a "time-domain" analysis: fields are defined at spacetime points, and the path integral sums over infinitely many histories — the QFT version of "reading the waveform," and the root of the measure problem and the divergences. Our spectral method is a "frequency-domain" analysis: we look not at spacetime points but at vibrational modes. Physical quantities change from "functions of spacetime points" into "sums of spectral eigenvalues"; the path integral becomes a discrete series, and "solving differential equations" becomes "summing a series."

But there is a fundamental inversion: in signal processing the time domain is “real” and the frequency domain is a tool; we turn this around — the spectrum is the reality, and spacetime is merely the “composite waveform” that all the modes superpose into. This is the mathematical landing point of Chapter 1’s conclusion (spacetime does not exist).

The inversion has a price: the hardest thing for mainstream physics to accept is causality and locality. Whether two events one light-year apart are causally connected is decided in the time domain by distance; in the spectral domain it must be re-derived — the “neighbour relations” of the spectrum must translate, in the emergent spacetime, into “spacelike separation.” This is the hardest bone of the whole method, and the part our papers address with the Osterwalder–Schrader reconstruction.

Dimension	Signal processing (time → frequency)	Our spectral method (coordinates → spectrum)
Original basis	Time domain (the time axis)	Coordinate space (spacetime points)
Transformed basis	Frequency domain (the frequency axis)	Spectral space (the vibrational modes of the internal space)
Core difficulty	Superposition of complex signals, noise filtering	The divergences of QFT, the nonexistent path-integral measure
Solution	Fourier transform: convolution becomes multiplication	Spectral sum: the path integral becomes a discrete series
Key operation	Fourier transform	Spectral decomposition / spectral sum
Physical meaning	“Wave” becomes “frequency”	“Field” becomes “vibrational modes (particle states)”

3. The Deep Water of the Yang–Mills Conjecture: What Is the Nature of the Field?

A million-dollar question

In 2000, the Clay Mathematics Institute announced seven "Millennium Prize Problems," each carrying a one-million-dollar prize. The first among them is the **Yang–Mills existence and mass gap problem**.

It asks two specific things:

1. **Existence:** can the quantum Yang–Mills theory in four-dimensional spacetime (the strong-force part of the Standard Model is precisely this) be rigorously, mathematically constructed without contradiction?

2. Mass gap: why do its elementary excitations (gluons) have mass — and specifically a "gap" mass, i.e. there exists a lowest non-zero mass, below which there is nothing?

More than twenty years on, these two questions still have no rigorous answer. Countless brilliant people have tried and none has fully cracked it.

The difficulty is stuck on "the nature of the field"

Why so hard? On the surface it looks like insufficient mathematical technique. But we hold that **the real difficulty is stuck on a philosophical question: what is a field, exactly?**

The traditional approach of quantum field theory goes like this:

1. First define the "field" — a mathematical object pervading spacetime (the **container** of change);
2. Then put "change" (particles, interactions) into it — let change happen inside this container;
3. Finally use the quantization rules (path integral, measure) to make this container "move."

Notice the order: **the container first, the change after**. The field is "the box that holds change."

Putting the cart before the horse: change is the substance

We hold that this order is **backwards**.

The correct order is reversed:

- **Change comes first; it is the substance (ontology);**
- **The field is the structure formed by change** — the relatively stable shape into which a great deal of change "freezes."

Use the river analogy: the traditional approach is "first the riverbed (field, container), then let the water (change) flow in it"; we say, **there is no riverbed at all — the water (change) flows, and as it flows it "carves out" a shape that looks like a riverbed; that shape is what we call the "field."**

An even better analogy: a crowd of people (change) walks back and forth; where many people walk, a "path" is trodden into the ground. **The path (field) is not there first; it is trodden out by the people (change)**. Traditional quantum field theory is "build the road first, then let people walk"; we say "people walk, and the road emerges on its own."

One can also use music: a melody (field) is, in essence, a string of notes (change). The traditional approach is "first the melody, then let the notes fill it"; we say, **it is the notes (change) whose combination constitutes the melody (field)**. The melody is not a container holding notes; the melody is itself one way of organizing the notes.

The ontological turn: from "things that exist" to "change itself"

This is the ontological turn of our entire method:

> **Not "change happens inside the field," but "the field is generated by change." Change is the substance; the field is derivative.**

Once we accept this, the two difficulties of the Yang–Mills conjecture are recast:

- "Does the quantum field exist?" becomes "Does change have a self-consistent, computable description?" (this is easier to answer: yes);
- "Where does the mass gap come from?" becomes "Why does the spectrum of change have a lowest non-zero frequency?" (this too is easier to answer: because the internal space is compact and finite, the spectrum has a gap by nature).

In other words: **we no longer try to "directly construct the field" (that is constructing the container of change, backwards); instead we start from "change" itself, letting the field emerge as the structure of change.**

This turn is the starting point of everything that follows. It repositions the century-old difficulty of "constructing quantum field theory" from a "problem of mathematical technique" into a "problem of the starting point" — the problem is not "how to build the container," but "we should never have started from the container at all."

What, then, does this "change" concretely amount to? Chapter 4 explains how to describe change with the spectrum; Chapter 6 gives the full answer — the smallest, most fundamental change is pinned down uniquely by a single axiom, which we call the disorder axiom.

The Yang–Mills conjecture: where exactly is the difficulty?

Yang–Mills theory is the mathematical skeleton of the strong force (QCD) in the Standard Model. The Clay Institute's million-dollar prize is really asking two very specific questions:

First, existence: can this quantum theory, in four-dimensional spacetime, be rigorously constructed without blowing up (no infinities)?

Second, the mass gap: why are gluons (the carriers of the strong force) forever "imprisoned" inside hadrons in experiment, so that physicists never see a free gluon? Mathematically, this means the theory's lowest excited state must have a **positive** energy (greater than zero), not a continuum starting from zero.

Why does the traditional method get stuck? Because traditional quantum field theory first erects a "spacetime stage," then defines the field fluctuations on that stage. And in four-dimensional spacetime, the "amplitude" of these fluctuations oscillates violently at short distances (high energy), so that the "mass gap" — that positive number — can never be pinned down: you get either zero (massless) or infinity.

Our answer: move the question from "four-dimensional spacetime" to "a three-dimensional compact internal space (RP^3)."

In a compact space, the vibrational modes are naturally discrete — like a string fixed at both ends, whose frequencies can only be the fundamental, the second harmonic, the third

harmonic, ... The lowest non-zero frequency is, by nature, a "gap." The mass gap m_Δ in our framework is **precisely the lowest non-zero frequency of this RP^3 "string."** It is not "pieced together" by complicated perturbative calculation; it is an **intrinsic property of the spectrum**.

So rather than saying we "solved" the Yang–Mills conjecture, it is more accurate to say we **bypassed its original setup**: we do not construct fields in four-dimensional spacetime; we define physical quantities on a three-dimensional compact spectrum — and the gap is simply already there.

> **In one sentence**: the difficulty of the Yang–Mills conjecture is that the "container" itself diverges; we replace the container with a compact spectrum, and the "gap" stops being a problem and becomes the starting point.

4. A Different Starting Point: From "Change," via the Spectral Sum

How to "grasp" change?

If change is the substance, then the first practical question is: **how do we mathematically "grasp" and analyze change?**

The answer: with the **spectrum**.

What is a spectrum? It is "a set of frequencies."

Anything that changes, whatever it is, can be decomposed into a set of elementary vibrational modes, each with a frequency. Listing all these frequencies gives the "spectrum" of the system.

- A string has its spectrum (fundamental + overtones);
- A drum has its spectrum (the various frequencies of its membrane);
- An atom has its spectrum (the transition frequencies between electron energy levels);
- Our internal space (RP^3) has its spectrum too (the frequencies of its various vibrational modes).

The spectrum is the most natural mathematical language of "change." You cannot say what "change" is as a "thing," but you can list all the frequencies of change completely — that is the most faithful, least embellished description of it.

Why is the spectrum more fundamental than the field? Because the field is "a quantity pervading spacetime," and it secretly depends on the (non-existent) background of spacetime; whereas the spectrum is merely a string of frequencies, **it needs no spacetime background — it needs only "change" itself**. You can define change inside an abstract internal space and ask for its spectrum — never once mentioning "spacetime."

The spectral sum: dually rebuilding everything from the spectrum

Our method is called the **spectral-sum representation**.

The idea: since change is the substance and the spectrum is the complete description of change, **we start from the spectrum and use it to "dually" rebuild everything physicists need — fields, geometry, gauge structure, particles, interactions.**

An analogy: a piece of music is, in essence, a string of notes (the spectrum). We do not assume "music must be housed in a container called a score"; we directly analyze the string of notes itself. And then we are astonished to find: **from this string of notes, harmony, melody, and rhythm "grow" on their own — the very things traditionally thought to require a "pre-existing score structure."**

This is the meaning of "duality": **the spectrum (change) and the physics (field, geometry, gauge) are two faces of one and the same thing.** Starting from the spectrum, you dually see the physics. You never have to assume the physics was "already there."

A concrete example, to feel how "the spectrum dually produces geometry": the shape of a drumhead (geometry) determines its timbre (spectrum); conversely, if you only listen to the drum (spectrum), you can infer the shape of the drumhead (geometry). **Spectrum and geometry mutually determine each other.** The traditional approach is "first the drumhead (geometry/spacetime), then the sound (spectrum)"; we reverse it: **start from the sound (spectrum), and the drumhead (geometry/spacetime) emerges as its dual.** This is the core insight of "from spectrum to spacetime."

One-sentence summary of our route

Traditional quantum field theory: **first build the field (container) → then put change in → then stitch it with quantization rules (measure).**

Our route: **change is the substance → describe change with the spectrum → dually emerge field, geometry, gauge from the spectrum → finally emerge the full physics of 4-dimensional spacetime.**

This route bypasses the "measure depends on spacetime" dead end: because we never start from "fields on spacetime," but from "the spectrum of change." The spectrum needs no "spacetime background" — it needs only "change" itself.

How do physicists judge whether a theory is "rigorous"? — the Wightman axioms

In mathematical physics, there is a recognized gold standard for judging whether a quantum field theory is "watertight": the **Wightman axioms**. Proposed by Wightman in the 1950s, they comprise several rules (roughly: a Hilbert space exists, a vacuum state exists, the field operators

are distributions, relativistic covariance holds, locality holds, the spectrum condition holds, and so on).

Any "serious" quantum field theory had better fit inside the Wightman framework. The Standard Model does — but proving rigorously that it satisfies the Wightman axioms in full is extremely hard, and no one has done it yet.

What is the relation between our framework and the Wightman axioms?

To be direct: **we do not try to "prove" that our framework satisfies the Wightman axioms.**

Why? Because the Wightman axioms presuppose "a fixed four-dimensional spacetime background" (Poincaré symmetry, the vacuum, and the Hilbert space are all defined on that background). Our starting point is "spacetime does not exist" — so the starting point is different.

But we did one thing: **in the "free-field" limit, our spectral-sum construction reduces to a free quantum field theory that satisfies the Wightman axioms rigorously** (this is the content of Appendix C of the paper). And in the interacting part, we switch language — using the "spectral sum" in place of the "measure," thereby bypassing the "spacetime-first" assumption on which the Wightman axioms rest.

This is not to say the Wightman axioms are "wrong"; rather: **they are an excellent map, but we are sailing waters that have no map.** The only way we can earn trust is to "touch shore at the known coast" (the free field).

> **In one sentence:** the Wightman axioms are the traffic rules of the standard road; we travel by water, writing our own rules — but we connect seamlessly to the road at the shore.

5. Several Key Points of the Method

This chapter breaks our method into a few operational technical points, each corresponding to a concrete computation already implemented in code.

Point 1: three levels of the spectrum

We distinguish three levels of the spectrum, from concrete to abstract:

- 1. Discrete spectrum** (most concrete): specific quantities one by one — particle masses, energy gaps, the hierarchy of three fermion generations, the gauge coupling constants. These are what finally get compared against observation.
- 2. Spectral sum** (middle level): summing over the spectrum, taking traces, computing densities — for example the "window capacity" (how many spectral modes a window can hold). This is the bridge from the discrete spectrum to the continuum limit.

3. Continuum limit / regularisation (most abstract): using the Weyl law, heat-kernel expansion, KK reduction, and so on, to handle the behavior of the spectrum at infinity. This is what makes the computation "convergent, non-divergent."

All three levels are indispensable: without the discrete spectrum there is nothing to compare against observation; without the spectral sum there is no internal self-consistency; without the continuum limit there is no "non-divergence." This three-layer structure corresponds to the complete chain "concrete $\leftrightarrow$ middle $\leftrightarrow$ abstract."

Point 2: two-end regularisation

The spectrum "diverges" at both ends, and each end needs separate handling:

- **Ultraviolet end (UV, high frequency)**: the higher the frequency, the more modes, and the sum explodes. We use a **Gaussian window** (heat kernel) to smoothly cut off the high-frequency modes — not a crude one-shot chop, but a gentle "closing of the window" with a Gaussian function. Why a Gaussian window rather than a hard cutoff? Because a hard cutoff leaves artificial "boundary marks," while the Gaussian window is natural, smooth, and leaves no artifacts.
- **Infrared end (IR, low frequency)**: the lower the frequency, the larger the reach involved, and the sum extends to infinity. We use the **principle of maximal entropy** to fix the infrared end — let the entropy be maximised under the constraints.

Only when both ends are handled is the middle spectral sum well-defined. This "two-end regularisation" is the cornerstone that makes the whole method "computably convergent."

Point 3: entropy is the bridge between UV and IR

One quantity that is unexpectedly present throughout is **entropy**.

We find that the "span" from the ultraviolet window to the infrared window, after taking its logarithm, is $\int \gamma_M = \ln(\text{window span}) = 139.253$. This number is exactly a Boltzmann entropy $S = \ln W$ — where W is "the number of ways of change the window can hold."

Entropy becomes the bridge connecting ultraviolet and infrared, connecting the microscopic and the macroscopic. This is the most elegant thread in the whole method: it ties the purely counting question "how many ways can change change" to the question "what does the universe look like."

Entropy here is not the "degree of disorder" of thermodynamics, but a more universal concept: **entropy = the logarithm of the number of ways of change.** How many different ways of change this window can hold — its logarithm is the entropy. And it is precisely this entropy that welds the two extremes, the high-frequency end (Planck scale) and the low-frequency end (Hubble scale), into one body.

Point 4: the highest principle — duality emergence

The highest principle running through everything, we call "**duality emergence**":

> spectrum → duality → gauge/geometry/entropy → emergence → 4D physics.

Four dualities are unified in one framework:

- **conformal** ↔ **gauge** (the scaling symmetry of spacetime dual to gauge symmetry);
- **geometry** ↔ **gauge** (the geometry of the internal space dual to the gauge-group structure);
- **UV** ↔ **IR** (high frequency dual to low frequency, bridged by entropy);
- **spectrum** ↔ **physics** (the frequencies of change dual to observable physical quantities).

"Duality" means: **the two sides are one and the same thing, seen from a different angle**. The reason we can rebuild everything from the spectrum is precisely that these dualities guarantee the equivalence of the two sides. The whole framework has no "arbitrary assumptions" — only these dualities interlocking and mutually confirming one another.

6. The Smallest Change, and the Uniqueness of Four Dimensions

Starting from "the smallest change"

Change comes in sizes. Some changes are complex, some simple. So: **what is the smallest, most fundamental change?**

Our answer is a one-sentence axiom, which we call the **disorder axiom**:

> **There exists a minimal, unbiased, spin-1/2 change.**

Unfolded, it has three clauses:

(A1) Existence: the change exists. This only says that our starting point is not "nothing," but change itself.

(A2) The quantum unit is spin-1/2: the carrier of the change is a two-dimensional complex space — the smallest unit of quantum mechanics. Together with "the symmetry group is minimal," this uniquely pins the symmetry group to $SU(2)$: a faithful continuous irreducible representation on a two-dimensional complex carrier can only have $SU(2)$ as its symmetry group (also known as $Spin(3)$).

(A3) Unbiasedness: the change plays no favourites with its symmetry group — there is no hidden preference. Together with the maximum-entropy principle, this uniquely gives a Gaussian distribution — a "structureless" random field (the scale field).

Simple as it sounds, this axiom is a derivation machine:

- spin-1/2 naturally distinguishes left from right (the chiral structure), and the chiral structure requires the symmetry group to be Spin(4);
- gluing antipodal points together ($z \mapsto -z$) turns the chiral structure into geometry: the internal space is $\mathbb{RP}^3 = S^3/Z_2$ — this is where $\mathbb{RP}^3$ comes from;
- “compact” means bounded and closed, like a drumhead, so the vibrational modes are discrete and finite. The “spectrum of the smallest change” is therefore the spectrum of $\mathbb{RP}^3$ — a discrete set of frequencies.

So the disorder axiom is the single starting point of the whole framework: a minimal, unbiased, spin-1/2 change. Everything that follows — the gauge group, the three fermion generations, the masses, the couplings — emerges from this axiom. Without it, $\mathbb{RP}^3$ would be just “some space picked by hand”; with it, $\mathbb{RP}^3$ is derived uniquely.

Starting from the spectrum of $\mathbb{RP}^3$ is therefore starting from a “smallest, most symmetric set of changes”: smallest, because it is one of the simplest closed three-dimensional spaces; most symmetric, because it inherits the sphere’s high symmetry, constrained by Z_2 (antipodal identification) in the simplest way.

Why four dimensions are "four"

A long-standing puzzle in physics: why is the spacetime we live in exactly **3+1 = 4 dimensions**? Why not 5, or 10?

In our framework, **four dimensions are not "chosen," but "uniquely self-consistent."** The skeleton of the derivation:

- the dimension d of the internal space and the colour number N_c (the "3" of the strong force SU(3)) are pinned by a duality equation: $d = N_c = 3$;
- this duality equation comes from the **conformal–gauge duality** $N_g \cdot \xi = 1$ — where $N_g = 8$ is the number of generators of the colour gauge group, and $\xi = 1/8$ is the conformal coupling constant, the two being reciprocals;
- this "reciprocal" equation uniquely gives $d = N_c = 3$ (it is the unique positive solution of $(N_c-3)(N_c+2) = 0$);
- the internal space is 3-dimensional; add 1 dimension of time, and you get **3+1 = 4-dimensional spacetime**.

In other words: **"four dimensions" is not an input assumption, but a derived result.** Change it, and the whole framework becomes inconsistent. The reason this result is clean is that it comes from a "reciprocal" duality relation — 8 and 1/8 multiply to 1, and this simple relation precisely, uniquely locks the dimension.

The lowest layer of the universe has the highest symmetry

Behind this lies a deep regularity:

> **The closer to the "smallest, most fundamental change," the higher the symmetry.**

Near the Planck scale (the smallest scale we can describe), symmetry is highest; moving toward the macroscopic, symmetry "breaks" more and more, and there is less of it. The electromagnetic, weak, and strong forces look distinct at low energy, but they all descend from one and the same higher symmetry at high energy.

"Four dimensions are unique" and "symmetry is highest" are two faces of the same thing: **at the bottom layer, change is simplest, most symmetric, and therefore most unique.** There is no room for "parameter tuning" — everything emerges necessarily from the spectrum of "the smallest change." This answers the deepest philosophical question: "why is the universe the way it is?" — because at the bottom layer it can only be this way: simplest, most symmetric, most unique.

7. Gravity, Gauge, Matter: The Trinity of Emergence

This chapter answers a core question: **what is the relationship among gravity, gauge forces, and matter?**

Gravity is emergent: transparent gravity

Chapter 1 already established the picture of "gravity emerging like temperature." Here we pin it into concrete conclusions.

In our framework, **gravity is not an independent, fundamental force, but is "emergent."**

Specifically, gravity corresponds to a special component of the spectrum — the **zero mode** (the mode of zero frequency, i.e. the "unchanging, most stable change"). We call it **"transparent gravity."**

This means three counter-intuitive things:

- 1. No curved spacetime:** there is no need to assume spacetime curves. Gravity's effects come from the zero mode of the spectrum, not from "the curvature of spacetime geometry."
- 2. No dark-matter particle:** there is no need to assume an invisible "dark-matter particle." Gravity's extra effects come from the spectral structure of the zero mode.
- 3. No ad hoc MOND correction:** there is no need to bolt an empirical "modified gravity" formula onto Newtonian gravity. The so-called MOND acceleration a_0 emerges naturally in the framework as $a_0 = c \cdot H_0 / (2\pi)$ — it is the infrared behavior of transparent gravity, not an external parameter.

In plain words: **gravity is not "the curvature of space," but "the most stable component of change."** Just as temperature is "the most averaged component of molecular motion" — the hot

water scalding you is the collective motion of molecules; the gravity you feel is the collective stable structure of vacuum fluctuations.

"Emergent gravity" was not our idea first. The reader may know that the Dutch physicist Erik Verlinde proposed, around 2010, a famous view: gravity is not a fundamental force but an **entropic force** — it arises from the distribution of information on a boundary, much as the pressure of a gas arises from molecular impacts. His slogan was: "gravity is a gradient of entropy."

Our "gravity = zero mode" and Verlinde's "gravity = entropy" resonate deeply — **both are the collective macroscopic behavior of microscopic degrees of freedom**. But we are more specific: Verlinde needs to assume a "holographic screen" and an information distribution on a boundary; we need no boundary — we simply pick out, directly from the **spectrum**, the "lowest-frequency, most stable" mode and identify it as gravity. Entropy, in our framework, is not the source of gravity but the **measure of the window span** (that 139.253). Gravity is the zero mode, entropy is the window capacity — two faces of one and the same spectrum.

To put it in parallel: Verlinde says "gravity is like temperature"; we say "gravity is like the most stable mercury column in a thermometer" — not the average of all the molecules, but that one **most unchanging structure**.

Gauge construction: gauge fields emerge from the spectrum

Similarly, the gauge fields (the carriers of the electromagnetic, weak, and strong forces) are not "basic inputs," but **emerge** from the spectrum of the internal space — specifically, the KK zero modes of the spectrum (the constant modes on the compact space).

The structure of the gauge group (why it is $SU(3) \times SU(2) \times U(1)$) is also derived from the spectrum, not assumed. The geometry of the internal space, through "duality," determines the algebra of the gauge group — this is the concrete meaning of the "geometry $\leftrightarrow$ gauge" duality.

The self-consistent cycle: gauge forms matter, matter deepens gravity

Thus a beautiful self-consistent cycle appears:

1. gravity (the spectral zero mode) **influences** the construction of gauge fields;
2. gauge fields **form** matter (fermions, the Higgs);
3. matter in turn **deepens** gravity (through the feedback of the spectrum).

This is not a "chicken or egg" dilemma, but a **self-consistently closed cycle** — every step is computed from the spectrum, and in the end they bite together into a whole. This is also the meaning of the word "closure" in our framework: **no quantity is bolted on; all quantities mutually determine one another inside the cycle**. Gravity "builds" gauge fields, gauge fields "build" matter, and matter "returns" to gravity — the three are three faces of one and the same spectrum.

There is no Big Bang

A striking corollary: **our framework has no "Big Bang" singularity.**

In traditional cosmology, the universe expands and redshifts, and tracing back into the past there is a "Big Bang" starting point — a singularity of infinite density where spacetime is torn apart. But in our framework, **redshift is not "the universe expanding," but "the spectrum tilting"** — redshift is a feature of the spectrum itself, not a stretching of space.

Since there is no "space expanding," there is no "starting point of expansion," and therefore **no Big Bang singularity.** The past of the universe is not a "point" but a "state" of the spectrum. The so-called "age of the universe," in the framework, is the logarithm of the window span of the spectrum, not "the time elapsed since a Big Bang."

Don't rush to believe it: a few honest caveats

A good science writer puts the objections on the table before the reader does. Here are the questions a reader is most likely to "throw at us."

Q: If spacetime doesn't exist, then what is "oscillating" in the gravitational waves that LIGO detected?

A: It is the zero mode of the vacuum-fluctuation spectrum that is oscillating. Just as a drumhead need not exist for the drum's sound (a vibrational mode) to be real — gravitational waves are not "ripples of spacetime" but "waves of the spectral zero mode." What we detect is the collective oscillation of the most stable mode in the spectrum.

Q: You say there is no Big Bang, so where does the cosmic microwave background (CMB) come from?

A: The CMB is not the "afterglow" of a Big Bang; it is the equilibrium radiation of the spectrum at the infrared end (the largest information boundary). It is a "floor," not the residue of a "history."

Q: Can this theory be falsified? If a theory explains everything, it explains nothing.

A: It can. We give concrete numerical predictions for 147 parameters (Chapter 8). If any one of them comes out wrong, beyond its stated precision, the theory collapses. It does not "say everything"; it "says a great many specific numbers, and waits to be tested."

Q: Aren't you just shoving everything you don't know into the word "spectrum," to dodge the question?

A: The spectrum is not a fig leaf — it is **computable**. Every term of the spectrum, every sum, has an explicit formula, and all of it can be run on a computer. It is hard, not vague.

Where are we now? — a conversation with the other "theories of everything"

On the frontier of physics, we are not the only ones trying to unify gravity and quantum mechanics. You have surely heard of **string theory**, **loop quantum gravity**, and the **holographic principle**. They are all big, and all famous. What is the relation between our framework and theirs? Are we a replacement, or a companion?

1. String theory: replace particles with vibrating strings

String theory says the elementary particles are not points but tiny closed or open "strings." Different vibrational modes correspond to different particles (one mode is the photon, another the graviton). Spacetime is the "stage" on which the strings exist, and it is not 4-dimensional but 10- or 11-dimensional — the extra dimensions are "compactified" (curled up), which is why we do not see them.

Our fundamental disagreement with string theory: string theory is "**spacetime first, vibration second**" — the vibrations (strings) exist inside spacetime; we are "**change (spectrum) first, spacetime second**" — spacetime emerges out of change. So string theory is "vibration inside a container"; we are "vibration giving rise to the container." String theory must assume 10 or 11 dimensions; we derive $3+1 = 4$.

But an interesting similarity: string theory also makes heavy use of "spectra" (the vibrational frequency spectra of strings), and so do we. The difference: string theory's spectrum is the spectrum of "**strings in spacetime**"; ours is the spectrum of "**the internal compact space**" itself. We take the "internal space" that string theory needs for compactification and make it the protagonist, while the string itself is "reduced" to the more fundamental modes of change on that space.

2. Loop quantum gravity: chop spacetime into little pieces

The idea of loop quantum gravity is: do not quantize gravity as a "graviton"; quantize "space" itself. Space is not continuous but made of discrete "loops," and area and volume have minimum units. It sidesteps the problem that "spacetime curvature needs a background," because it drops the background altogether and works directly with quantized geometry.

Our kinship with loop quantum gravity: we are both "**background-independent**" theories. Neither of us believes spacetime is pre-existing. But the divergence is this: loop quantum gravity still treats "discrete geometry" as the fundamental substance ("space is made of little loops"), whereas we treat "**the spectrum of change**" as the fundamental substance — geometry merely emerges from the spectrum; it is not primordial. In other words, loop quantum gravity says "the bricks of the world are space quanta"; we say "the bricks of the world are frequencies."

3. The holographic principle: compress the universe onto a boundary

The holographic principle comes from black-hole thermodynamics and a branch of string theory (the AdS/CFT correspondence). It says all the physical information in a spatial region can be completely encoded on the "boundary" of that region. The universe you thought was three-dimensional is really the projection of a two-dimensional "hologram."

Our resonance with the holographic principle: our "spectrum $\rightarrow$ spacetime" is also a **duality**. The holographic principle is the duality "boundary $\leftrightarrow$ bulk"; ours is the duality "spectrum $\leftrightarrow$ spacetime." Both are "information before geometry." The difference: the holographic boundary is usually infinite and needs a fixed background; our spectrum is compact and needs no background. Our duality is more "thorough" — it needs no boundary at all, just a string of frequencies.

A comparison table:

Theory	Fundamental substance	Where spacetime comes from	Dimensions
String theory	vibrating strings	pre-existing (a stage)	10/11 (assumed)
Loop quantum gravity	space quanta (loops)	quantized geometry directly	4 (quantized)
Holographic principle	fields on the boundary	projected from the boundary duality	4 (dual)
Our framework	the spectrum of change	emerges from the spectrum	4 (derived)

We do not claim the other theories are "wrong." We only say: **they all still need some pre-existing "something" (strings, loops, a boundary), whereas we have done away with that "something" too, leaving only "change itself."**

Coincidence or necessity? — what the 2026 Fields Medals tell us

In 2026, two Chinese mathematicians received the Fields Medal at the same time.
Deng's work proves: **microscopic particles can emerge into macroscopic temperature, pressure, and fluids**. Wang's work reveals: **continuous space can, at microscopic scales, show an astonishing, almost-discrete complexity**.

One looks upward from below (discrete $\rightarrow$ continuous); the other looks downward from above (continuous $\rightarrow$ discrete). They are like two mirrors reflecting each other.

And our framework stands at their crossing point: **if the microscopic can emerge into the macroscopic (Deng), and if the continuous can expose the complexity of the discrete (Wang), then why cannot "spacetime" itself emerge from something more fundamental — change?**

Behind this there is an even grander story — **Hilbert's sixth problem**.

In 1900, the mathematician David Hilbert posed 23 problems that would point the direction for all of twentieth-century mathematics. The sixth is special: it is not a concrete conjecture but a

grand program — **"the axiomatisation of physics."** In plain terms: can we, as Euclid did for geometry, find a minimal set of axioms for all of physics, and then derive every physical law from them?

One of its concrete versions is the axiomatisation of fluid mechanics: microscopically, a single molecule obeys Newtonian mechanics (reversible, no direction of time); macroscopically, a gas made of countless molecules exhibits temperature, pressure, and viscosity, and obeys the second law of thermodynamics (irreversible, with an arrow of time). **But the logical chain from the microscopic to the macroscopic had never been rigorously proved — it had been missing for 125 years.**

Deng and his collaborators filled in that missing link: starting from Newton's hard-sphere model (microscopic), they rigorously obtained the Boltzmann equation (mesoscopic statistics), and then derived the fluid-mechanics equations (macroscopic). **This erected, between the microscopic and the macroscopic, a logically airtight bridge that was previously thought impossible.** This is the solution of the "narrow Hilbert sixth problem."

A romantic way to put it: **Deng recovered the arrow of time for Newtonian mechanics.** The motion of a single molecule has no direction, but countless molecules, through statistical regularity, "grow" an irreversible arrow of time — the most striking embodiment of "emergence."

Now, bring our gaze back to our own framework.

Deng proved with mathematics that "emergence" is not philosophical rhetoric but a rigorous fact: **microscopic particles can emerge into macroscopic fluids, temperature, and the arrow of time.**

And our framework is asking a deeper question: **if temperature is emergent, if fluids are emergent, if the arrow of time is emergent, then is "spacetime" itself also emergent from something more fundamental?**

Our answer is: **yes. That more fundamental thing is "change."** We do not need to assume the "stage" of spacetime exists first — starting directly from "the spectrum of change," spacetime, gravity, gauge fields, and matter all "grow" out of "change."

If Hilbert's sixth problem is **"finding axioms for the known physical world,"** then our framework asks a more fundamental question: **"before the 'physical world' appears, what is the axiom?"**

Our answer is: **change. The spectrum of change is the axiom, and all of physics is its corollary.**

Deng put the final full stop on "how the microscopic emerges into the macroscopic"; our framework is writing the opening line of "how change emerges into spacetime." **This, in one word, is an axiom that solves everything.**

8. The World That Was Computed: A Survey of Results

This chapter gives hard results. Everything before was "ideas"; here is "what came out of the computation."

One anchor, 146 derived

The whole framework has **147 parameters**. Of these:

- **1 observational anchor:** Newton's gravitational constant G_N — the only quantity "borrowed" from experiment, used solely for the final comparison and never entering any computation.
- **146 derived quantities:** everything else — gauge couplings, Yukawa couplings, the Higgs vacuum expectation value, quark/lepton masses, cosmological parameters, the QCD scale — **all derived from the internal spectrum, zero hard-coding, zero fitting.**

There is no "parameter tuning." You cannot alter any internal quantity in order to match an observed value. Whatever the deviation is, it is reported as-is. This is precisely where the method is honest: it does not "fit" the data; it "predicts" the data, then lets the deviations speak for themselves.

A survey of precision

How well the computed results agree with observation:

Quantity	Precision
Gauge coupling constants	< 0.01%
Electroweak scale (Higgs v)	< 0.01%
Fermion mass ratios	< 1%
Cosmological fractions (Ω)	< 1%
QCD scale	~1%

This precision means: this "start from the spectrum, zero free parameters" method can reproduce most of the observed values of the Standard Model and cosmology — not by tuning parameters, but by structure itself.

A few striking concrete results

- **Exact flatness:** $\Omega_b + \Omega_{DM} + \Omega_\Lambda = 1.00000$ — the matter density + dark-matter density + dark-energy density of the universe add up to exactly 1 (a flat universe), not fitted but an identity.
- **The dark-energy density** $\Omega_\Lambda = 2/3 + r_{23}/(3\pi) = 0.68504$, consistent with observation.
- **The Hubble constant**, the **cosmic microwave background temperature**, and the **proton mass** are all computed from internal quantities.

- **Three fermion generations:** why exactly three? It comes from the "window capacity" of the RP^3 spinor spectrum — the window $(kL)^2$ holds exactly three spinor modes.
- **Neutrinos, flavour mixing, CP violation:** these also emerge from the geometry of the spectrum.

How it is verified

All results are reproduced in one go by a master-chain script:

```
py scripts/reproduce_v4.py          # all 40 chain items pass
py scripts/audit_param_writers.py # every parameter has a provenance audit, CLEAN
```

There are also 17 Lean 4 machine-proof files, which verify one by one, with a theorem prover, the framework's core "content identities" (the integer relations that determine the physics). That is: the key symmetry relations of the framework have not only numerical verification but also **machine-checkable formal proof**.

Why bring up code in a popular-science piece? Because too many "beautiful theories" in the history of physics ended up stuck on paper, unable to produce concrete numbers. The 40 chain items here are **something a computer can run through in one night and then tell you "dark-energy density = 0.685."** And the Lean 4 theorem prover is a machine that checks for you whether "the logic contradicts itself." **This is not experimental verification, but it is a mathematical "structural inspection."** The numbers tell you "does it match observation"; the proofs tell you "is there a hole in the logic" — two rulers, each measuring one face.

9. Asking After the Essence: Entanglement, Information Boundaries, Dark Energy, Dark Matter, Spacetime, Gravity

This final chapter uses the language of the framework to answer, one by one, the most "philosophical" questions. These are all natural corollaries of the framework, not explanations pasted on afterward.

What is the essence of entanglement?

Traditionally, quantum entanglement is described as a mysterious "action at a distance" — two particles, even far apart, instantly "know" each other's state.

In our framework, **the essence of entanglement is "the correlation of the spectrum"**: two particles are entangled because they share one and the same mode in one and the same spectrum. They are not "acting at a distance across space"; rather, **they belong to the same mode of change from the outset** — the concept of "spatial distance" is itself what their shared spectrum "projects out." Entanglement precedes space; it does not happen inside space.

A unifying view: everything is a "boundary of information"

Before answering about horizons, dark energy, and dark matter, first establish a unifying view — it will string these seemingly unrelated questions onto one thread.

The boundary of information: a system can hold, transmit, and access only a finite amount of information. The boundaries of these limits are the essence of our various "scales."

Under this view:

- **the Planck scale = the smallest boundary of information** (the limit of information resolution);
- **the Hubble scale = the largest boundary of information** (the limit of information reach);
- **the black-hole horizon = the gravitational boundary of information** (the limit beyond which information cannot escape).

The three, seemingly unrelated, are actually one and the same thing — **the three boundaries of "information": the smallest, the largest, and the place where gravity is strong enough to "seal off" information.**

The black hole: the gravitational boundary of information

What is a black hole? In the framework, it is not the mysterious object "where spacetime is infinitely curved," but **the boundary of gravity — the place where information is "sealed off."**

Specifically: when mass (energy) gathers to a sufficient degree, it begins to **affect the vacuum fluctuations, so that the vacuum fluctuations can no longer "maintain" their stability.** The critical point of this "fluctuation destabilisation" is the black-hole horizon. Inside the horizon, the stable structure of the vacuum fluctuations is broken, and information can no longer pass out in the usual way; outside the horizon, everything proceeds as normal.

So the black-hole horizon is not "a singularity of spacetime," but **the boundary where "gravity is strong enough to destabilise the vacuum fluctuations."** It is an "information boundary": information inside and outside is separated by this boundary. This also gives the framework's answer to the black-hole information paradox — not "information vanishes," but "information is sealed off by the boundary where the vacuum fluctuations destabilise."

The Planck scale and the Hubble scale: the smallest and largest information boundaries

The Planck scale and the Hubble scale are the two endpoints of the "information boundary" thread:

- **The Planck scale** (about 10^{-35} m): **the smallest information boundary.** Below this scale, change itself can no longer be resolved — this is the smallest unit of "change," and the absolute lower limit of "information resolution." In the framework it corresponds to the ultraviolet Gaussian window — the window edge of the high-frequency spectral modes.

- **The Hubble scale** (about 10^{26} m): **the largest information boundary**. Beyond this scale, changes can no longer influence one another — this is the largest reach of mutual influence of "change," and the absolute upper limit of "information access." In the framework it corresponds to the infrared entropy maximum — the window edge of the low-frequency spectral modes.

One smallest, one largest. In between lies the entire world we can perceive, measure, and talk about. **The universe is the "window" between these two information boundaries.** And the logarithm of the span of this window is precisely the entropy **139.253** from Chapter 5 — it measures how many ways of change this "window of information" can hold.

What is the essence of dark energy?

Dark energy (the "thing" driving the accelerating expansion of the universe) is, in the framework: **the neutrino floor.**

Specifically, $\rho_\Lambda = Y_u \cdot m_{\nu 1}^4$ — the dark-energy density equals the fourth power of the lightest neutrino mass, times a Yukawa coupling. It is not a mysterious "fifth element," but **the "floor" of the lightest-neutrino spectral mode at the infrared end.** The accelerating expansion of the universe is, in essence, the behavior of the spectrum at this floor.

In other words, dark energy is not "extra energy added in," but "the intrinsic behavior of the spectrum at the largest information boundary (the Hubble scale)."

What is the essence of dark matter?

Dark matter (the extra, invisible gravity in galaxies) is, in the framework: **the TT spectral zero mode of transparent gravity.**

It needs no "invisible particle." The extra gravity in galaxy rotation curves comes from the structure of the spectral zero mode — this is exactly what the name "transparent gravity" means: **gravity is the zero mode of the spectrum; it exists "transparently," transmitted by no particle at all.** So dark matter is not "a particle we cannot find," but "a misreading of the spectral essence of gravity." We mistake "the extra gravitational effect of the spectral zero mode" for "an invisible particle," when in fact there is no dark-matter particle at all.

What is the essence of spacetime?

Returning to the opening question, we can now give the complete answer:

> **Spacetime does not "exist." It is "the description of change" — time is the ordering of change, space is the reach of change.**

It is neither the "curved entity" of General Relativity nor the "rigid background" of the Standard Model. It is merely an **ordered structure** we read out of "the spectrum of change." Just as temperature is the emergence of molecular motion, spacetime is the emergence of change —

except that temperature is the emergence of "average kinetic energy," while spacetime is the emergence of "ordering and reach."

What is the essence of gravity?

> **Gravity is the zero mode of the spectrum — the most stable, most unchanging part of "change."**

It is not a "force," not "the curvature of space," and it needs no mysterious particle called a "graviton" transmitting in between. Gravity is simply the quietest undertone of change itself. Just as temperature is "the average of molecular motion," gravity is "the stability of vacuum fluctuations" — both are emergence, both are "the collective macroscopic behavior of microscopic degrees of freedom."

What is this theory good for?

To be honest, this theory does not build phones or generate electricity. It is still far from any technical application — farther than science fiction.

But it offers a new **way of thinking**, a new handle different from "curved spacetime": **if gravity is emergent, then perhaps someday we could manipulate gravity by "changing the spectrum of the vacuum fluctuations."** Today this sounds like fantasy, but two hundred years ago "transmitting information with electromagnetic waves" sounded like fantasy too.

More importantly, it answers the question "why still do basic science": **not to be useful immediately, but to understand what kind of world we actually stand in.** If even "spacetime" and "gravity" are not the most fundamental, then our understanding of ourselves still has an enormous, unopened space.

Conclusion: The Central Idea in One Sentence

Compress the whole thread into one sentence:

> Change is the substance, the spectrum is the language of change, and the field, geometry, gauge, gravity, matter — even spacetime itself — are all structures that emerge from the spectrum of the smallest change fixed by the disorder axiom (a minimal, unbiased, spin-1/2 change).

The entire familiar universe is not an edifice "already there," but a "river of change" — the water flows on its own, carving out the riverbed (spacetime), the whirlpools (particles), the undercurrents (dark energy, dark matter), and also marking out its boundaries (the Planck scale, the black-hole horizon, the Hubble scale).

The deepest, quietest stretch of this river (the zero mode of the spectrum) is what we call "gravity" — like temperature, it is not a fundamental thing, but an emergent phenomenon. And its source is the smallest, most symmetric, most unique of all changes.

Change is everything.

If change is the substance, then what are "you"? — You are not a fixed, unchanging entity; you are an **extremely complex, temporary, stable spectral structure** formed by the countless changes inside your body (heartbeat, breath, brainwaves, cell metabolism). So, **you yourself are the emergence of change.**

This may be the gentlest gift this framework offers to ordinary people: you are not a tiny, lonely speck of dust in the universe; you are a single, unique wave, blooming right now in the great river of "change."

Appendix: a tiny glossary

If you forget a word later on, come back here — you will remember it at a glance.

- **Spectrum:** the "frequency list" of change. Like the list of pitches of piano keys.
- **Zero mode:** that special mode with frequency 0 — "unchanging change"; it is the essence of gravity.
- **Window capacity:** how many elementary vibrational modes a finite space can hold at most.
- **Heat kernel / Gaussian window:** a smooth pair of "scissors" that gently cuts off overly high-frequency vibrations, to stop the calculation from blowing up.
- **KK zero mode:** the "constant mode" on a compact space — a vibration that does not vary with position; it is the origin of gauge fields.
- **Duality:** two faces of one thing, seen from a different angle. For example "spectrum" and "geometry" are a pair.
- **Emergence:** collective behavior that cannot be found microscopically but truly exists macroscopically. Like temperature relative to molecules.
- **Disorder axiom:** the single starting point of the framework — "there exists a minimal, unbiased, spin-1/2 change."
- **Changing the analysis domain:** moving a problem from the "spacetime coordinate" basis to the "vibrational mode (spectrum)" basis, like moving a time-domain problem to the frequency domain in signal processing.

This document is a popular science introduction to the two papers (I. Gauge structure and fermion content; II. Effective couplings and mass scales) and the "Coarse-Graining Genesis

Framework V4.0," intended to help grasp the central ideas; it does not replace the rigorous derivations in the papers and the code. The rigorous form of the "gravity-emerges-like-temperature" and "information boundary" pictures, and all numerical conclusions, are subject to the machine verification in `scripts/reproduce\_v4.py`, `scripts/audit\_param\_writers.py`, and `lean\_proofs/`, and to Paper I.